Lección 8

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Plan de trabajo

Formulación Lagrangiana

– Aplicaciones:

* movimiento en campo central, dispersión de partículas
* observadores no inerciales

► Nuestro Syllabus:

Nuestro Syllabus:

FIS-310:

UNIDAD 2. Formulación Lagrangiana
2.1. Variables Generalizadas
2.2. Clasificación de los Sistemas Mecánicos
2.3. Elementos de Cálculo Variacional
2.4. Ecuaciones de Lagrange
2.5. Aplicaciones: movimiento en un campo central, observadores no inerciales

UNIDAD 3. Teoremas de Conservación
3.1. Teorema de Noether.
3.2. Aplicaciones del teorema de Noether.

FIS-345:

Unidad I: Principios Variacionales, Lagrangianos y Hamiltonianos

- Principio de Hamilton y Lagrangianos
- Simetrías y Leyes de Conservación (Teorema de Noether)

Unidad II: Algunas Aplicaciones

- Movimiento en campo de Fuerzas Centrales
- Movimiento de Sólido Rígido

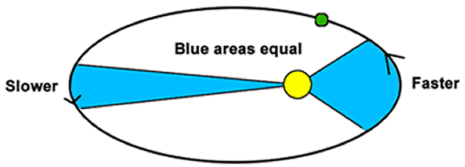
FIS-426:

1) Principios Variacionales, Lagrangianos y Hamiltonianos

- Principio de Hamilton y Lagrangianos
- Hamiltonianos
- Otros principios variacionales
- Simetrías y Leyes de Conservación (Teorema de Noether)

Summary of the last lecture: Motion in Central field

- Motion always occurs in a plane, and the second law of Kepler is valid for any potential: a line segment joining a planet and the Sun sweeps out equal areas during equal intervals of time



Integrals of motion:

- Angular momentum (absolute value):

$$M^2 = m^2 r^4 \left(\dot{\theta}^2 + \sin^2\theta\,\dot{\varphi}^2\right) = \mathrm{const} \qquad (1)$$

- Energy:

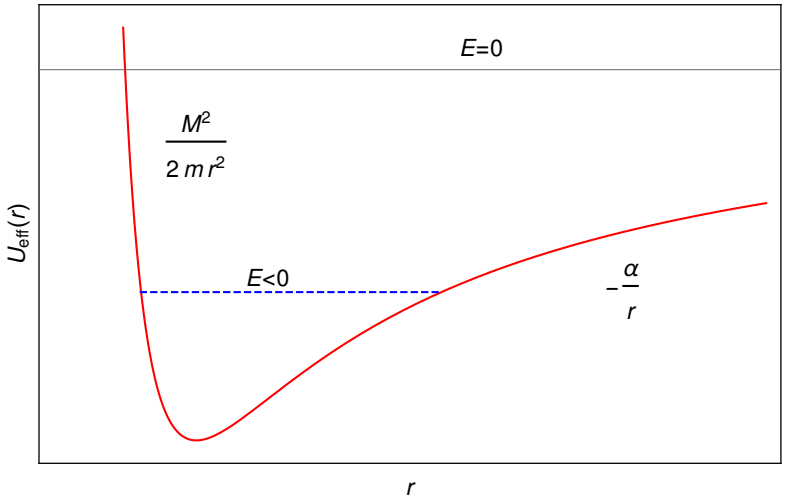
$$E = \frac{m\left(\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\,\dot{\varphi}^2\right)}{2} + U(r) = \frac{m\dot{r}^2}{2} + \underbrace{U(r) + \frac{M^2}{2mr^2}}_{U_{\mathrm{eff}}(r)} \qquad (2)$$

$$\int^r \frac{dr}{\sqrt{\frac{2}{m}\left(E - U_{\mathrm{eff}}(r)\right)}} = t - t_0 \quad \phi(t) = \int^t dt\,\frac{M}{m\,r^2(t)} \qquad (1)$$

$$\phi - \phi_0 = \int^r dr\,\frac{M}{m\,r^2\sqrt{\frac{2}{m}\left(E - U_{\mathrm{eff}}(r)\right)}} \qquad (2)$$

Motion in central field - Kepler's problem

- Special case: the field of Coulomb $U(r) = -\frac{\alpha}{r}$ (assume that $\alpha > 0$, attractive potential)



- Case $E > 0$: particle scatters on center (goes to infinity)
- Case $E < 0$: particle is bound (moves between $r_{\min}$ and $r_{\max}$)
- For $M \neq 0$ the particle cannot fall on the center
- For $\alpha < 0$ (repulsion) $U_{\text{eff}}(r) > 0$ and slowly decreases, only scattering

Motion in central field - Kepler's problem

Case $E < 0$: particle is bound (moves between $r_{\min}$ and $r_{\max}$ along the elliptical orbit, $0 \leq e < 1$)

$$\phi - \phi_0 = \ldots = \arccos\left(\frac{\frac{M}{r} - \frac{m\alpha}{M}}{\sqrt{2mE + m^2\alpha^2/M^2}}\right)$$

$$\frac{p}{r} = 1 + e\,\cos\phi$$

$$p = \frac{M^2}{m\alpha}, \quad e = \sqrt{1 + \frac{2EM^2}{m\alpha^2}}$$

Kepler's first law

The orbit of a planet is an ellipse with the Sun at one of the two foci.

Case $E = 0$: $e = 1$ (parabola)
Case $E > 0$ (scattering): $e > 1$ (hyperbola)

Motion in the Coulomb field - time dependence

- Up to now we considered the shape of the orbit; the evaluation of the temporal dependence $r(t)$ is straightforward:

$$t - t_0 = \int^r \frac{dr}{\sqrt{\frac{2}{m}\left(E - U_{\text{eff}}(r)\right)}} = \sqrt{\frac{m}{2|E|}} \int^r \frac{r\,dr}{\sqrt{\left(-r^2 + \frac{\alpha}{|E|} r - \frac{M^2}{2m\,|E|}\right)}}$$

$$= \sqrt{\frac{ma}{\alpha}} \int^r \frac{r\,dr}{\sqrt{(a^2 e^2 - (r-a)^2)}}$$

Let's change integration variable:

$$r = a - a\,e\,\cos\xi$$

$$\Rightarrow t - t_0 = a\sqrt{\frac{ma}{\alpha}} \int^r \frac{\cancel{a\,e\,\sin\xi}(1 - e\,\cos\xi)\,d\xi}{\cancel{a\,e\,\sin\xi}} =$$

$$= a\sqrt{\frac{ma}{\alpha}}\,(\xi - e\,\sin\xi)\,, \quad \xi = \arccos\left(\frac{r-a}{a\,e}\right)$$

-parametric expression for the trajectory (Landau, Chapter 15 for details)

Kepler's problem - 3rd Kepler's law

$$M = mr^2 d\phi/dt = \mathrm{const}$$

$$MT = M\int dt = m\int r^2 d\phi = m\,\pi\,a\,b 2\pi a^{3/2}\sqrt{m/\alpha}$$

$$a = \frac{p}{1-e^2} = \frac{\alpha}{2\,|E|}, \quad b = \frac{p}{\sqrt{1-e^2}} = \frac{M}{\sqrt{2m\,|E|}}$$

- Gravitational field: $\alpha \equiv G\,m\,m_{\mathrm{Sun}} \Rightarrow \sqrt{m/\alpha} = 1/\sqrt{G\,m_{\mathrm{Sun}}}$, same for all planets

$$\Rightarrow T^2 \sim a^3$$

Kepler's third law

The square of the orbital period of a planet is proportional to the cube of the semi-major axis of its orbit.

- Consistent with **mechanical similarity** law: if $\boldsymbol{r}(t)$ is trajectory, then

$$\forall\alpha, \quad \tilde{\boldsymbol{r}}(t) = \alpha\,\boldsymbol{r}\left(\alpha^{3/2}t\right)$$

is also a trajectory.

Periodicity

$$\frac{p}{r} = 1 + e\,\cos\phi$$

Observation:

In the Coulomb field **all** bound trajectories are closed: particle returns back to initial position with the same velocity $\boldsymbol{v}$ and starts motion over the same trajectory

Caution

The orbits are "closed" (periodic) only for a few (2) potentials.

–In general potential $U(r)$ *some* orbits may be closed, but others NOT

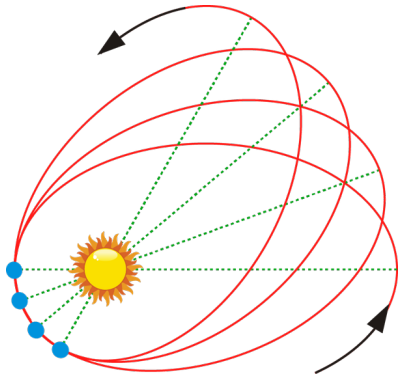
Example:

–The Coulomb's potential with minor perturbations,

$$U(r) = -\frac{\alpha}{r^{1+\gamma}}, \qquad \text{or} \qquad -\frac{\alpha}{r} + \frac{\gamma}{r^n} \quad |\gamma| \ll 1, \quad n > 0$$

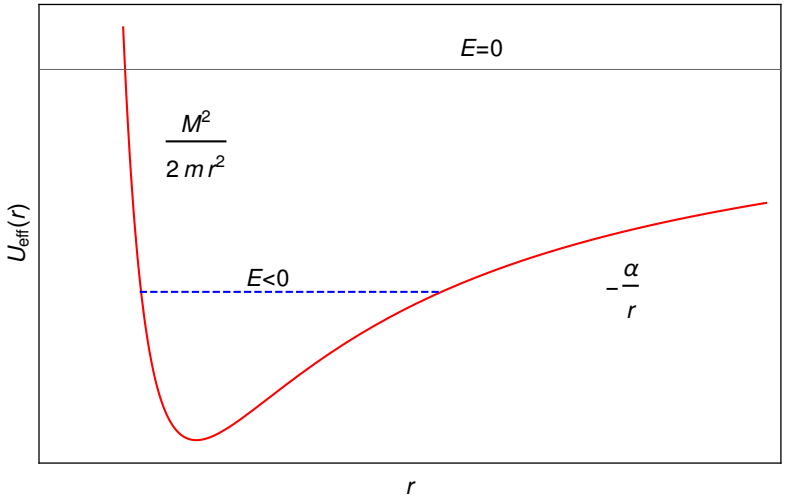
–Circular orbits are still closed, just circumferences

–For non-circular orbits have precession, see the plot



Closed orbits

- **Our goal**: understand when orbits are *closed* in arbitrary potential $U(r)$.
  - We do NOT impose condition that ALL orbits are closed, we only try to find those specal cases when they are closed



Lemma

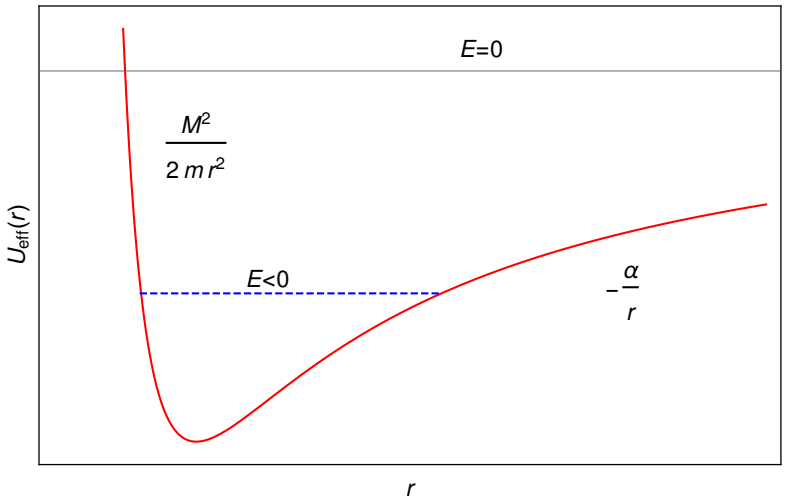
All orbits with given angular momentum $\boldsymbol{M}$ and energy $E = U_{\text{eff}}(r_{\text{min}})$, where r_{min} is solution of

$$-\frac{M^2}{m r_{r_{\text{min}}}^3} + U'(r_{\text{min}}) = 0, \qquad (1)$$

are closed

Please suggest how we can prove this statement ???

Closed orbits



All orbits with given angular momentum $\boldsymbol{M}$ and energy $E = U_{\text{eff}}(r_{\text{min}})$, where r_{min} is solution of

$$-\frac{M^2}{m r_{r_{\text{min}}}^3} + U'(r_{\text{min}}) = 0, \qquad (1)$$

are closed:

(1) is trivial condition that centrif. force equals attraction force

►Radial equation of motion: $\ddot{r} = 0$;

$$r = r_{\text{min}} = \text{const}, \ \dot{\phi} = \frac{M^2}{2 m r_{\text{min}}^2}$$

−Particle moves along circular orbit with constant angular velocity

Lemma

Claim 2

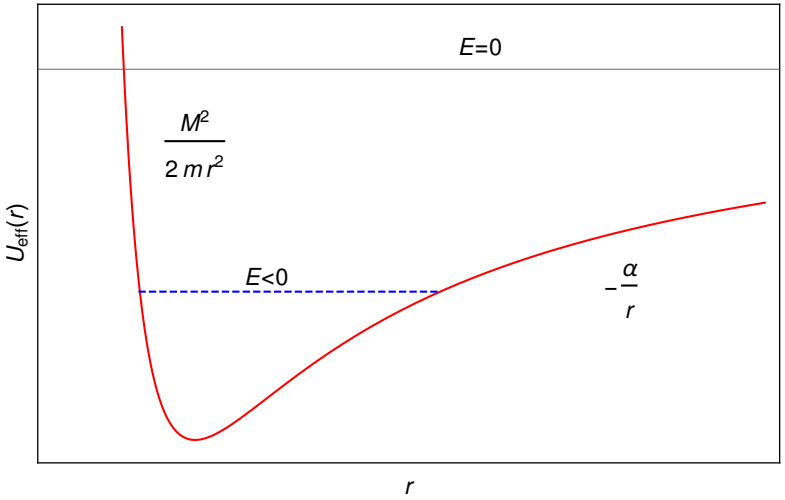
For orbits with given angular momentum $\boldsymbol{M}$ and energy $E = U_{\text{eff}}\left(r_{\text{min}}\right) + \varepsilon$, where ε is small correction, the orbits are closed if

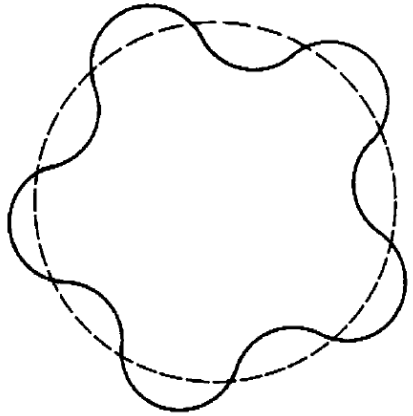
$$\frac{\partial^2 U_{\text{eff}}\left(r_{\text{min}}\right)}{\partial r^2} = \frac{\partial^2 U\left(r_{\text{min}}\right)}{\partial r^2} + \frac{3M^2}{m r_{\text{min}}^4} > 0$$

and parameter β defined as

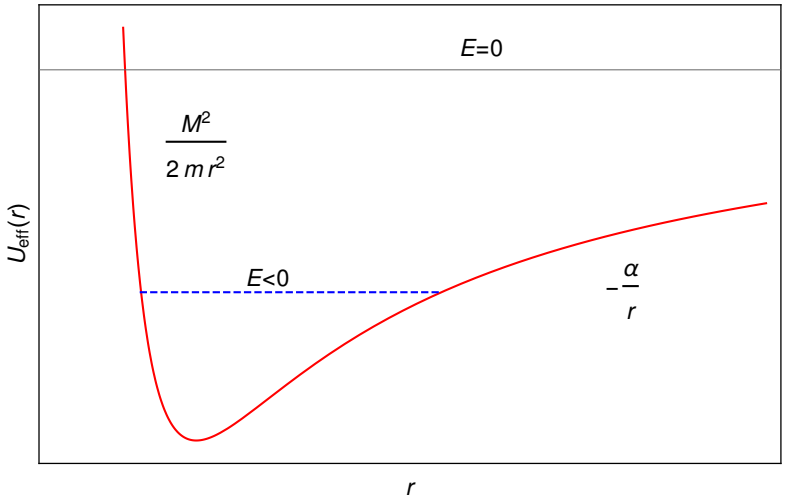
$$\beta^2 = -3 + \frac{r_{\text{min}}}{U'\left(r_{\text{min}}\right)} U''\left(r_{\text{min}}\right)$$

takes rational values, $\beta = n_1/n_2$ where $n_1, n_{2s} \in \mathbb{N}$





Proof



- For small energy $E = E_{\min} + \varepsilon$ may approximate the effective potential with parabola, so condition

$$\frac{\partial^2 U_{\text{eff}}(r_{\min})}{\partial r^2} = \frac{\partial^2 U(r_{\min})}{\partial r^2} + \frac{3M^2}{mr_{\min}^4} > 0$$

is just condition of stability (otherwise circular orbit $r = r_{\min}$ is unstable)

- Define $\xi = r - r_{\min} \ll r_{\min}$
  - radial equation of motion:

$$m\ddot{\xi} = -\kappa\xi \Rightarrow \xi = \xi_0 \cos\left(t\sqrt{\frac{\kappa}{m}} + \phi_0\right), \qquad \kappa := \frac{\partial^2 U_{\text{eff}}(r_{\min})}{\partial r^2}$$

  - angular equation of motion:

$$\dot{\phi} = \frac{M}{m(r_{\min} + \xi)^2} \approx \frac{M}{mr_{\min}^2}$$

Proof

$$m\ddot{\xi} = -\kappa\,\xi \Rightarrow \xi = \xi_0 \cos\left(t\sqrt{\frac{\kappa}{m}} + \phi_0\right), \qquad \kappa := \frac{\partial^2 U_{\text{eff}}\left(r_{\min}\right)}{\partial r^2}$$

$$\dot{\phi} = \frac{M}{m\left(r_{\min} + \xi\right)^2} \approx \frac{M}{m r_{\min}^2}$$

- Motion is periodic only if periods of radial and angular motion are conmensurate, i.e.

$$\sqrt{\frac{\kappa}{m}} \Big/ \left(\frac{M}{m r_{\min}^2}\right) \equiv \beta = \frac{n_1}{n_2},\ n_1, n_2 \in \mathbb{N}$$

- Some simplifications:

$$\beta^2 = \frac{\kappa/m}{M^2/\left(m r_{\min}^2\right)^2} = 3 + \frac{U''\left(r_{\min}\right)}{M^2/\left(m r_{\min}^2\right)^2}$$

We may take into account that in equilibrium point $U'\left(r_{\min}\right) = M^2/\left(m r_{r_{\min}}^3\right)$ to eliminate M^2 in denominator; this yields

$$\beta^2 = 3 + \frac{r_{\min}}{U'\left(r_{\min}\right)} U''\left(r_{\min}\right) = 3 + \left.\frac{d\ln f(r)}{d\ln r}\right|_{r=r_{\min}}, \quad f(r) = U'(r).$$

Closed orbits

If we introduce radial force $f(r) = U'(r)$, then

$$\beta^2 = 3 + \left.\frac{d \ln f(r)}{d \ln r}\right|_{r=r_{\min}}, \quad f(r) = U'(r).$$

Earlier we have seen that orbits are closed if $\beta = n_1/n_2$

$$\frac{d \ln f(r)}{d \ln r} = \beta^2 - 3$$

$$\ln f = \left(\beta^2 - 3\right) \ln r + \text{const}$$

$$\Rightarrow f(r) = \frac{\text{const}}{r^{3-\beta^2}} = \frac{\text{const}}{r^{3-(n_1/n_2)^2}}$$

$$\Rightarrow U(r) = -\frac{\text{const}}{r^{2-\beta^2}} = -\frac{\text{const}}{r^{2-(n_1/n_2)^2}}$$

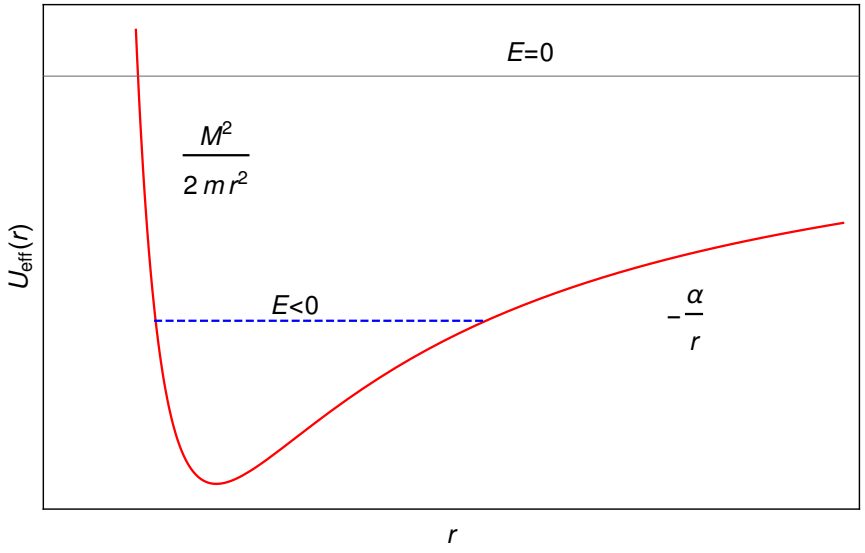
Thus periodicity requires that potential MUST have power law r-dependence

Closed orbits

- So far approximated the effective potential with parabola near $r = r_{\min}$,

$$U_{\text{eff}}(r) \approx U_{\text{eff}}(r_{\min}) + \\ + \frac{\partial^2 U_{\text{eff}}(r_{\min})}{\partial r^2} \frac{(r - r_{\min})^2}{2}$$

  – We can take into account higher order anharmonic terms
  – In general break periodicity condition

- 

Bernhardt theorem

Only for $\beta = 1$ (Coulomb) and $\beta = 2$ (3D harmonic oscillator) the orbits remain periodic for <u>any</u> energy

- Can check by explicit evaluation that in these potentials all the orbits are periodic

Motion in central field (short summary)

$$\int^{r} \frac{dr}{\sqrt{\frac{2}{m}\left(E - U_{\text{eff}}\left(r\right)\right)}} = t - t_0 \quad (1)$$

$$\phi(t) = \int^{t} dt\, \frac{M}{m\, r^2(t)}. \quad (2)$$

$$\phi - \phi_0 = \int^{r} dr\, \frac{M}{m\, r^2 \sqrt{\frac{2}{m}\left(E - U_{\text{eff}}\left(r\right)\right)}} \quad (3)$$

- Results (1-3) are *formal* solutions
  - allow to reconstruct numerically the trajectory.
  - analytical solutions known for $U(r) = \frac{\alpha}{r^n}$
    - *in terms of elementary functions for $n = 1, 2, -2$
    - **Case $n = 1$: Kepler problem: trajectories are conic sections (ellipse, parabola or hyperbola)
    - *in terms of elliptic functions for $n = -1, 3, \pm 4, \pm 6$

Cetral field - Inverse problems

Inverse Problems

Up to now we considered problems in which we have known potential and need to find trajectory

Now we'll see slightly different type of problems: we know trajectory (its properties) and need to find potential $U(r)$

Problems of this type: inverse problems

(Inverse) Problem 1

We analyze a motion of a pointlike mass m in unknown central potential $U(r)$. Using experimental observations of circular orbits $R = \mathrm{const}$, we found how the energy E depends on R. Using this information, find the potential $U(r)$.

Please suggest algorithm of solution

(Inverse) Problem 1 (solution)

For circular orbit $r = R = \mathrm{const}$ the particle does not have radial velocity, $v_r = 0$, so

$$E = \frac{mv_\phi^2}{2} + U(r), \quad M = mv_\phi r$$

Also, circular orbit corresponds to minimum of the $U_{\mathrm{eff}}(r)$, which may be reinterpreted as equality of centrifugal and (attractive) potential forces

$$-\frac{M^2}{mR^3} + U'(R) = 0, \qquad \Rightarrow \qquad -2(E - U(R)) + RU'(R) = 0,$$

Change variable $\zeta = \ln R$, so $R\,U'(R) = dU/d\zeta$

$$\frac{dU(\zeta)}{d\zeta} + 2U = 2E(\zeta),$$

How to solve this ODE?

Problem 1 (solution)

$$\frac{dU(\zeta)}{d\zeta} + 2U = 2E(\zeta),$$

First-order linear inhomogeneous ODE, the "force" in the right-hand side is known

$-$Change variables: $U(\zeta) = u(\zeta)e^{-2\zeta}$

$$\frac{du(\zeta)}{d\zeta} = 2E(\zeta)e^{2\zeta}$$

$$u = \int d\zeta\, 2E(\zeta)e^{2\zeta}$$

$$U = e^{-2\zeta}\int d\zeta\, 2E(\zeta)e^{2\zeta} = \frac{2}{R^2}\int \frac{dR}{R}\, E(R)R^2 = \frac{2}{R^2}\int dR\; E(R)R$$

$-$Case $E = \lambda\, R^n$:

$$U(R) = \frac{2\,\lambda}{n+2}\; R^n = \frac{2E(R)}{n+2}$$

in agreement with virial theorem

Problem 2

We analyze motion of a pointlike mass m in unknown central potential $U(r)$. Using experimental observations of circular orbits $R = \mathrm{const}$, we found how the period of motion T depends on R. Using this information, find $U(r)$. Demonstrate that for $T \sim R^{3/2}$ your result agrees with the 3rd law of Kepler.

Please suggest algorithm of solution

Problem 2 (reply)

For circular orbit $r = R = \mathrm{const}$ the particle does not have radial velocity, $v_r = 0$, so

$$E = \frac{m v_\phi^2}{2} + U(r), \quad M = m v_\phi r$$

Also, circular orbit corresponds to minimum of the $U_{\mathrm{eff}}(r)$, which may be reinterpreted as equality of centrifugal and (attractive) potential forces

$$-\frac{M^2}{mR^3} + U'(R) = 0, \qquad \Rightarrow \qquad -2(E - U(R)) + RU'(R) = 0,$$

Period of motion:

$$T = \frac{2\pi}{\omega} = \frac{2\pi R}{v_\phi} = \frac{2\pi R}{\sqrt{\frac{2}{m}(E - U(R))}} = \frac{2\pi R}{\sqrt{\frac{1}{m} R U'(R)}}$$

So conclude that

$$U = \int dR \, \frac{4\pi^2 m R}{T^2(R)}$$

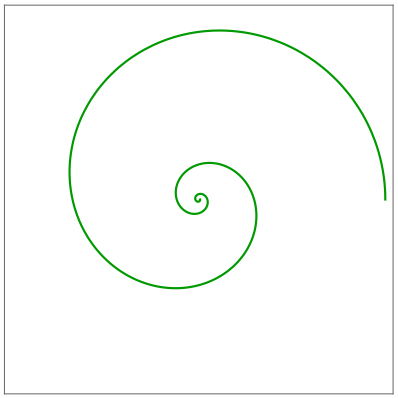
–Kepler problem: $T^2 = \kappa R^3$, so $U(R) = \left(4\pi^2 m/\kappa\right)/R$

–Harm. Oscillator: $T = \mathrm{const}$, $U(R) = \left(2\pi^2 m/T^2\right) R^2 = \left(m\omega^2/2\right) R^2$

Capture of particles

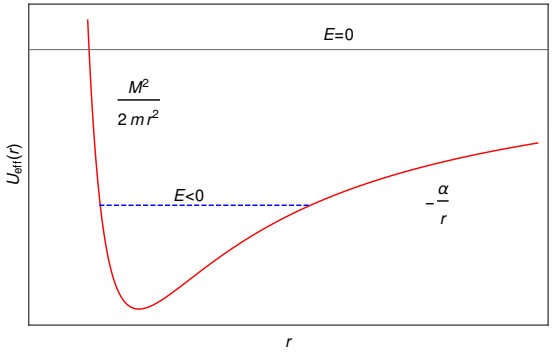
Problem 3

- So far we *assumed* that the particle does not fall on center (no capture).
- We understand that if potential is attractive and very strong, eventually particle will get caught (fall on center):



Try to formulate the condition which should satisfy a potential in order to have this behaviour.

- Recall that radial motion is 1D motion in effective potential.



Motion in the field $\sim 1/r^2$

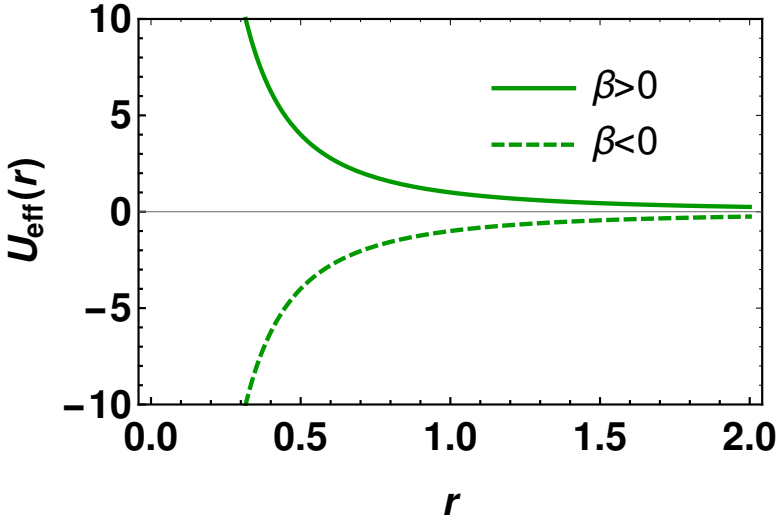
–Assume a particle of mass m moves in the central field $U(r) = -\frac{\alpha}{r^2}$, where $\alpha > 0$.

–

$$U_{\rm eff}(r) = U(r) + \frac{M^2}{2\,m\,r^2} = \frac{\beta}{r^2}$$

$$\beta = \frac{M^2}{2m} - \alpha$$

- Should distinguish cases $\beta > 0$ and $\beta < 0$



►In case $\beta > 0$ particle never falls on center ($r = 0$), centrifugal barrier

–We may have $\beta > 0$ even if $E < 0$

–At given E, value of M is directly related to angle between $\boldsymbol{r}$ and $\boldsymbol{v}$, $M = m\,r\,v\,\sin\theta_{rv}$

– M is maximal if $\boldsymbol{r}$ & $\boldsymbol{v}$ are perpendicular

– if $\boldsymbol{r}$ & $\boldsymbol{v}$ are collinear, $M = 0$

►In case $\beta < 0$ particle falls on center, scattering center captures the particle

–Maximal (critical) value of M when this happens: $M_{\rm crit} = \sqrt{2\,m\,\alpha}$

Motion in the field $\sim 1/r^2$

Trajectory:

$$\phi - \phi_0 = \int dr \, \frac{\frac{M}{m\,r^2}}{\sqrt{\frac{2}{m}\left(E - \frac{\beta}{r^2}\right)}},$$

Change variable: $r \to \xi = 1/r$

$$\phi - \phi_0 = M\sqrt{\frac{m}{2E}} \int \frac{d\xi}{\sqrt{1-\gamma\,\xi^2}}, \quad \gamma = \frac{\beta}{E}$$

Use

$$\int \frac{dz}{\sqrt{1-z^2}} = \arcsin(z), \quad \int \frac{dz}{\sqrt{z^2-1}} = \operatorname{arccosh}(z), \quad \int \frac{dz}{\sqrt{1+z^2}} = \operatorname{arcsinh}(z)$$

to take the integral or `Integrate[..., r]` of *Mathematica* with option *Assumptions*$\to$ $\{\ldots\}$; final result (see §15 in Landau):

(a) for $E > 0$ and $M^2/2m > \alpha$, $\dfrac{1}{r} = \sqrt{\dfrac{2mE}{M^2 - 2m\alpha}} \cos\left[\phi\sqrt{\left(1 - \dfrac{2m\alpha}{M^2}\right)}\right]$,

(b) for $E > 0$ and $M^2/2m < \alpha$, $\dfrac{1}{r} = \sqrt{\dfrac{2mE}{2m\alpha - M^2}} \sinh\left[\phi\sqrt{\left(\dfrac{2m\alpha}{M^2} - 1\right)}\right]$,

(c) for $E < 0$ and $M^2/2m < \alpha$, $\dfrac{1}{r} = \sqrt{\dfrac{2m|E|}{2m\alpha - M^2}} \cosh\left[\phi\sqrt{\left(\dfrac{2m\alpha}{M^2} - 1\right)}\right]$.

In all three cases

Motion in the field $\sim 1/r^2$

To get time dependence, we use

$$t - t_0 = \int^r \frac{dr}{\sqrt{\frac{2}{m}\left(E - U_{\text{eff}}\left(r\right)\right)}} = \int^r \frac{dr}{\sqrt{\frac{2}{m}\left(E - \frac{\beta}{r^2}\right)}}$$

$$= \int^r \frac{r\,dr}{\sqrt{\frac{2}{m}\left(E\,r^2 - \beta\right)}} = \frac{m}{2E}\sqrt{\frac{2}{m}\left(E\,r^2 - \beta\right)}$$

So we may find explicitly the t-dependence of the trajectory

$$\frac{4E^2}{m^2}\left(t - t_0\right)^2 = \frac{2}{m}\left(E\,r^2 - \beta\right)$$

$$r = \sqrt{\frac{E}{m}\left(t - t_0\right)^2 + \frac{\beta}{E}}$$

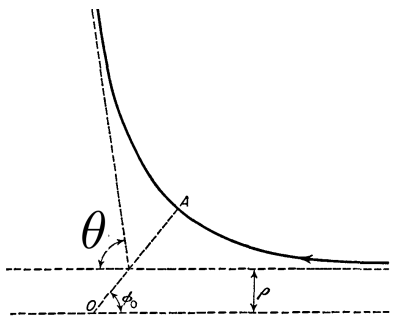
Motion in the field $\sim 1/r^2$: qualitative analysis

Case $E > 0$ and $M^2 > 2m\alpha$ ($\beta > 0$):

$$\frac{1}{r} = \sqrt{\frac{2mE}{M^2 - 2m\alpha}} \cos\left[(\varphi - \varphi_0)\sqrt{1 - \frac{2m\alpha}{M^2}}\right], \quad r(t) = \sqrt{\frac{2E(t-t_0)^2}{m} + \left(\frac{M^2}{2mE} - \frac{\alpha}{E}\right)}$$

Since energy E is positive, at infinity kinetic energy $T \neq 0$

– The solution is just a scattering, similar to Coulomb field



Since $r \geq 0$ (distance from center is always positive), conclude that particle may stay only in the region

$$|\varphi - \varphi_0| \leq \frac{\pi}{2} \Big/ \sqrt{1 - \frac{2m\alpha}{M^2}} \qquad (3)$$

trajectory is similar to hyperbolic motion in the Coulomb field

Motion in the field $\sim 1/r^2$

Case $E > 0$ and $M^2 > 2m\alpha$ ($\beta > 0$):

$$\frac{1}{r} = \sqrt{\frac{2mE}{M^2 - 2m\alpha}} \cos\left[(\varphi - \varphi_0)\sqrt{1 - \frac{2m\alpha}{M^2}}\right], \quad r(t) = \sqrt{\frac{2E(t-t_0)^2}{m} + \left(\frac{M^2}{2mE} - \frac{\alpha}{E}\right)}$$

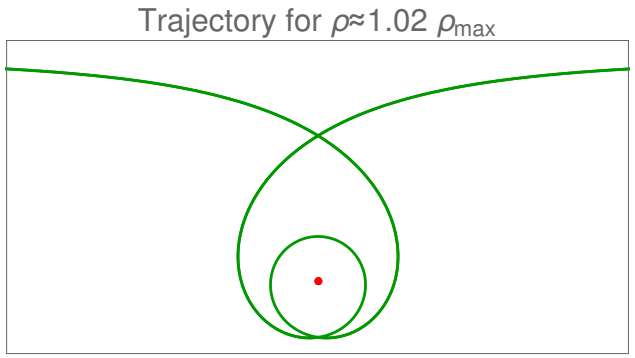
$$|\varphi - \varphi_0| \leq \frac{\pi}{2} \Big/ \sqrt{1 - \frac{2m\alpha}{M^2}} \tag{4}$$

For small values

$$\sqrt{1 - \frac{2m\alpha}{M^2}} \leq \frac{1}{4}$$

formally (4) implies that $\varphi - \varphi_0$ spans a region $(0, 2\pi)$. Physically this corresponds to a motion in which particle starts rotating around the center before escaping (see left).

Note however that cos(...) eventually becomes negative for some φ, so particle can't remain rotating forever ...



$$\rho \equiv M/mv_\infty, \ \rho_{\max} = \sqrt{2\alpha/mv_\infty^2}$$

Trajectory generated with Mathematica , using `PolarPlot[r[ϕ], {ϕ, 0, 6π}]`

Motion in the field $\sim 1/r^2$: qualitative analysis

Case $E > 0$ and $M^2 < 2m\alpha$ ($\beta < 0$):

$$\frac{1}{r} = \sqrt{\frac{2mE}{2m\alpha - M^2}} \sinh\left[(\varphi - \varphi_0)\sqrt{\frac{2m\alpha}{M^2} - 1}\right], \quad r(t) = \sqrt{\frac{2E(t-t_0)^2}{m} + \left(\frac{M^2}{2mE} - \frac{\alpha}{E}\right)}$$

Since energy E is positive, at infinity kinetic energy $T \neq 0$

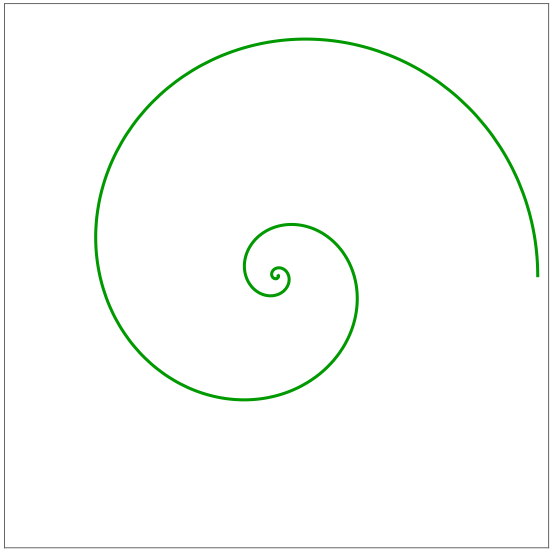
- Since $r \geq 0$ (distance from center is always positive), conclude that particle may stay only in the region

$$\varphi - \varphi_0 \geq 0$$

  Trajectory is a spiral line (see left), particle eventually falls to the center.
- From

$$r^2(t) = \frac{2E(t-t_0)^2}{m} + \frac{M^2}{2mE}\left(1 - \frac{2m\alpha}{M^2}\right) = 0$$

  can find the time it takes to fall to center.



Trajectory generated with Mathematica , using `PolarPlot[r[ϕ], {ϕ, 0, 6π}]`

Motion in the field $\sim 1/r^2$: qualitative analysis

Case $E < 0$ and $M^2 < 2m\alpha$ ($\beta < 0$):

$$\frac{1}{r} = \sqrt{\frac{2m|E|}{2m\alpha - M^2}} \cosh\left[(\varphi - \varphi_0)\sqrt{\frac{2m\alpha}{M^2} - 1}\right], \quad r(t) = \sqrt{\frac{2E(t-t_0)^2}{m} + \left(\frac{M^2}{2mE} - \frac{\alpha}{E}\right)}$$

Since energy E is negative, particle can't go to infinity: $U(\infty) = 0$ and kinetic energy always $T > 0$

There is no restriction on variable $\varphi - \varphi_0$. But as $\varphi - \varphi_0 \to \pm\infty$, can see that $r(\varphi)$ vanishes rapidly, particle falls on center. Effectively, particle is trapped (can't escape)

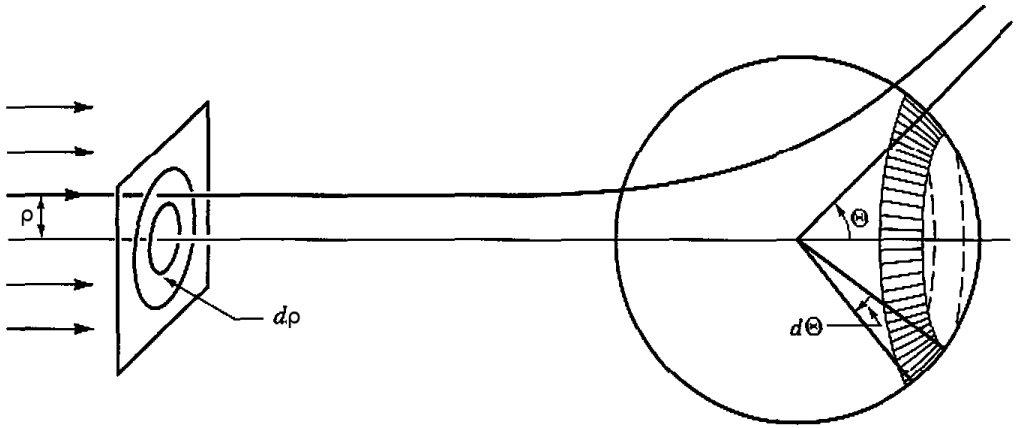
Scattering problem

Scattering in the external field

- Up to now we evaluated the shape of the orbit assuming that the initial conditions $(\boldsymbol{r}(0), \boldsymbol{v}(0))$ allow us to fix energy E, angular momentum $\boldsymbol{M}$ and the parameters of the orbit.

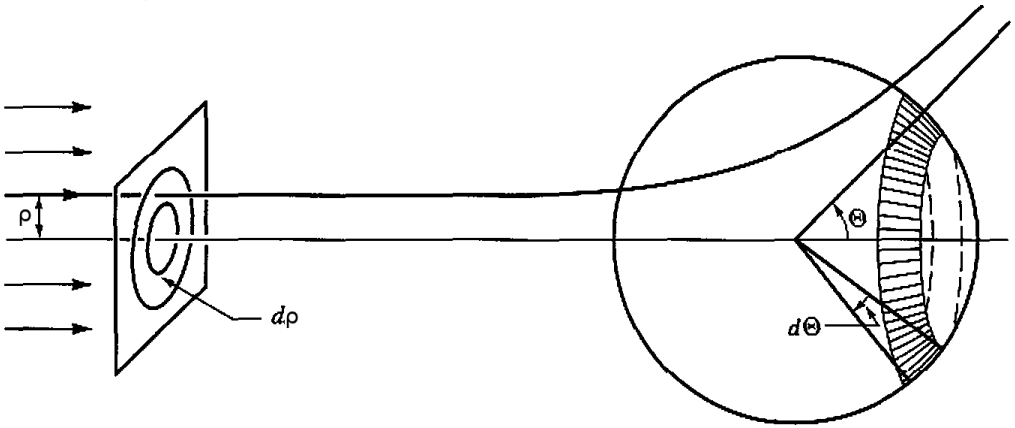
- In some problems we may now only part of this information:

−For example, we scatter a flux of particles with given energy E in the external field, and we need to find the *angular distribution* of particles after scattering. We do not know exact position $\boldsymbol{r}(0)$ to reconstruct the full trajectory of each individual particle



We expect that all particles which scatter with the same impact parameter ρ should have the same scattering angle $\Theta(\rho)$

Scattering in the external field



►

Flux of the particles J

Number of particles which pass through the unit area in transverse direction per unit of time

$$dN = J\,dS\,dt, \qquad J = \frac{dN}{dS\,dt}$$

In what follows we assume that flux J is *homogeneous*, *i.e.* does not depend on transverse coordinate

Differential cross-section

The number of particles scattered in a given solid angle $d\Omega$ or range of energies of scattered particles* $(E', E' + dE)$ and divided by the flux

$$\frac{d\sigma}{d\,\Omega} = \frac{1}{J}\frac{dN_{\text{scattered}}}{d\,\Omega} = \frac{1}{J}\frac{dN_{\text{scattered}}}{\sin\theta\,d\theta\,d\phi}$$

Scattering in the external field

- (Numerical value of) differential cross-section depends on choice of variables.

–If the two variables q and ξ are related to each other, then

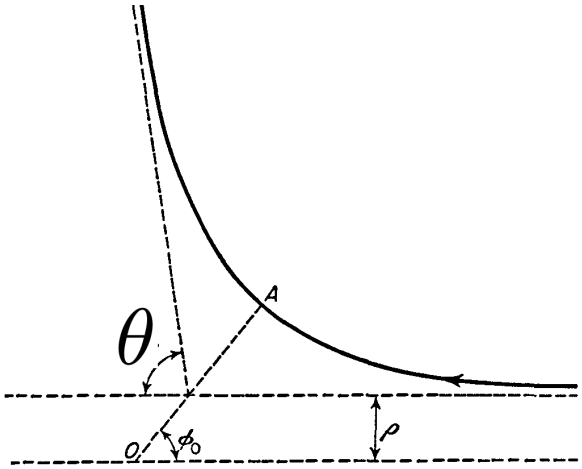
$$\frac{d\sigma}{dq} = \frac{d\sigma}{d\xi}\left|\frac{d\xi}{dq}\right|$$

–We may define differential cross-section using any generalized variables, not only angles or energies. The cross-section changes when we change variables as

$$\frac{d\sigma}{dq_1 \ldots dq_n} = \frac{d\sigma}{d\xi_1 \ldots d\xi_n}\left|\det\left(\frac{d\xi_a}{dq_b}\right)\right|$$

where $\det(\ldots)$ is a conventional Jacobian which describes change of the volume when we change coordinates. In case of just 1 variable, this is just chain rule.

Scattering in the external field

- 

  - Assume ϕ_0 is the angle between the one of the asymptotes and direction towards the closet point A, "periapsis" (in case of planets rotating around the sun it is aslo called perihelion, in case of other stars: periastron)
  - Both asymptote form the same angle ϕ_0 with direction towards
  - From geometry we may conclude that

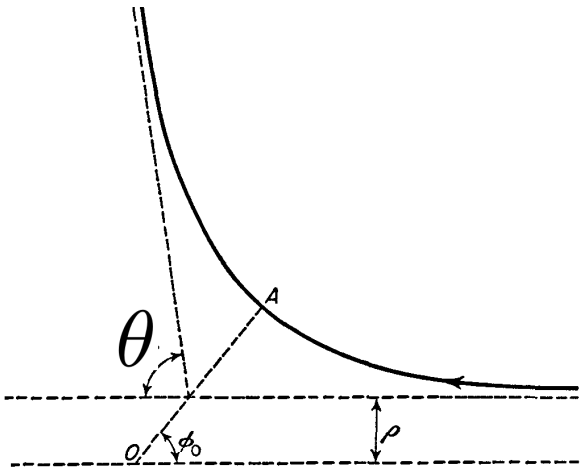
$$\theta = \pi - 2\phi_0$$

- Earlier we found the equation of trajectory in any potential in the form of indefinite integral. Adding now appropriate limits, we may get

$$\phi_0 = \int_{r_{\min}}^{\infty} \frac{\frac{M}{m\,r^2}\,dr}{\sqrt{\frac{2}{m}\left(E - U(r) - \frac{M^2}{2\,m\,r^2}\right)}} = \int_{r_{\min}}^{\infty} \frac{\frac{M}{m\,r^2}\,dr}{\sqrt{\frac{2}{m}\left(E - U_{\text{eff}}(r)\right)}}, \quad U_{\text{eff}}(r) = U(r) + \frac{M^2}{2\,m\,r^2}$$

  - The value $r_{\min}$ (distance to periapsis) may be found from condition that $E = U_{\text{eff}}\left(r_{\min}\right)$, it is return point for radial motion

Scattering in the external field



$$\frac{d\sigma}{d\theta} = 2\pi\rho(\theta)\left|\frac{d\rho}{d\theta}\right|, \quad \theta = \pi - 2\phi_0$$

$$\phi_0 = \int_{r_{\min}}^{\infty} \frac{\frac{M}{m\,r^2}\,dr}{\sqrt{\frac{2}{m}\left(E - U_{\text{eff}}(r)\right)}}, \quad U_{\text{eff}}(r) = U(r) + \frac{M^2}{2\,m\,r^2}$$

and $r_{\min}$ is found from

$$E = U_{\text{eff}}(r_{\min}) = U(r_{\min}) + \frac{M^2}{2\,m\,r_{\min}^2}$$

We assume that at infinity where $U(r) \approx 0$, all particles in the incident beam have the same velocity v_∞ and move in horizontal direction, so may fix the unknown parameters E, M from values at ∞:

$$E = \frac{m\,v_\infty^2}{2}, \quad M = |\boldsymbol{r} \times \boldsymbol{p}| = mv_\infty\,|\boldsymbol{r} \times \hat{\boldsymbol{x}}| = mv_\infty\rho\,|\hat{\boldsymbol{y}} \times \hat{\boldsymbol{x}}| = m\,v_\infty\,\rho,$$

$$\Rightarrow \phi_0 = \int_{r_{\min}}^{\infty} \frac{\frac{v\rho}{r^2}\,dr}{\sqrt{\frac{2}{m}\left(\frac{mv_\infty^2}{2} - U_{\text{eff}}(r)\right)}} = \int_{r_{\min}}^{\infty} \frac{\frac{v\rho}{r^2}\,dr}{\sqrt{\frac{2}{m}\left(\frac{mv_\infty^2}{2}\left(1 - \frac{\rho^2}{r^2}\right) - U(r)\right)}},$$

$$U_{\text{eff}}(r) = U(r) + \frac{mv_\infty^2}{2}\,\frac{\rho^2}{r^2}.$$

Example: Scattering in the Coulomb field

Rutherford's formula (CM frame)

In the previous lecture we analyzed in detail motion of a particle in the Coulomb field $U = -\alpha/r$. Using those results, find the cross-section of elastic scattering

$$\frac{d\sigma}{d\Omega} = \frac{1}{2\pi \sin\theta} \frac{d\sigma}{d\theta}$$

Reminder: trajectory in the Coulomb field:

$$\frac{p}{r} = 1 + e \cos\phi$$

- Case $E > 0$ (scattering): $e > 1$, trajectory is hyperbola with semiaxes

$$a = \frac{p}{e^2 - 1}, \quad b = \frac{p}{\sqrt{e^2 - 1}}$$

$$p = \frac{M^2}{m\alpha} = \frac{m v_\infty^2 \rho^2}{\alpha}, \quad e = \sqrt{1 + \frac{2EM^2}{m\alpha^2}} = \sqrt{1 + \left(\frac{m v_\infty^2 \rho}{\alpha}\right)^2}$$

Scattering in the Coulomb field

Minimal distance $r_{\min}$ which a particle may achieve:

$$E + \frac{\alpha}{r_{\min}} - \frac{M^2}{2\,m\,r_{\min}^2} = 0 \Rightarrow r_{\min} = \frac{-\alpha + \sqrt{\alpha^2 + \frac{EM^2}{2m}}}{2E} = \frac{\rho^2}{p}\,(e-1)$$

$$\phi_0 = \int_{r_{\min}}^{\infty} \frac{\frac{M}{m\,r^2}\,dr}{\sqrt{\frac{2}{m}\left(E + \frac{\alpha}{r} - \frac{M^2}{2\,m\,r^2}\right)}} = \arccos\left(\frac{1}{e}\left(\frac{p}{r_{\min}} - 1\right)\right) - \arccos\left(-\frac{1}{e}\right) =$$

$$= \arccos\left(\frac{\alpha/mv_\infty^2\rho}{\sqrt{1 + (\alpha/mv_\infty^2\rho)^2}}\right)$$

$$\Rightarrow \rho^2 = \left(\frac{\alpha}{mv_\infty^2}\right)^2 \tan^2\phi_0 = \left(\frac{\alpha}{mv_\infty^2}\right)^2 \cot^2\frac{\theta}{2} \quad \theta = \pi - 2\phi_0,$$

$$\frac{d\sigma}{d\Omega} = \frac{d\sigma}{2\pi\sin\theta\,d\theta} = \frac{1}{2\sin\theta}\left|\frac{d\rho^2}{d\theta}\right| = \left(\frac{\alpha}{2mv_\infty^2}\right)^2 \frac{1}{\sin^4(\theta/2)}$$

(Rutherford's formula)

Scattering in the Coulomb field

$$\frac{d\sigma}{d\Omega} = \left(\frac{\alpha}{2mv_\infty^2}\right)^2 \frac{1}{\sin^4(\theta/2)}$$

- Note that

$$\sigma_{\text{tot. elast}} = \int d\Omega \, \frac{d\sigma}{d\Omega} = \infty$$

-divergence due to singular behaviour at $\theta \approx 0$ ($\rho \approx \infty$).

$\Rightarrow$this is a consequence of long-range Coulomb interaction. In any other potential which falls faster at infinity, the total scattering cross-section remains finite,

$$\sigma_{\text{tot. elast}} = \text{finite}.$$

Control question

Rutherford formula (scattering in external field):

$$\frac{d\sigma}{d\Omega} = \left(\frac{q_1 q_2}{2\mu v_\infty^2}\right)^2 \frac{1}{\sin^4(\theta/2)}$$

In all our previous considerations we assumed that flux J is homogeneous, does not depend on impact parameter ρ.

Assume that You have a beam in which the flux is not homogeneous,

$$J = J(\rho) = J_0 \exp\left(-\frac{\rho^2}{\rho_0^2}\right), \qquad J_0 = \mathrm{const}$$

but You do not know this and measured experimentally the cross-section as

$$\frac{d\sigma}{d\,\Omega_0} = \frac{1}{J_0}\frac{d\mathcal{N}_{\mathrm{scattered}}}{d\,\Omega}$$

What will be the distribution of particles which You will detect in the laboratory?

Control question (reply)

In the cross-section $d\sigma = 2\pi\rho d\rho$ the number of particles which passes through the ring per unit of time is

$$d\mathcal{N}_{\text{scattered}} = J(\rho)\, 2\pi\rho d\rho \qquad \text{instead of} \quad J_0\, 2\pi\rho d\rho$$

so in all expressions for the cross-section we should add the factor $J(\rho)/J_0$, expressed in terms of the angle θ:

$$\rho^2 = \left(\frac{\alpha}{mv_\infty^2}\right)^2 \cot^2\frac{\theta}{2}$$

so

$$J(\rho)/J_0 = \exp\left(-\left(\frac{\alpha}{mv_\infty^2\rho_0}\right)^2 \cot^2\frac{\theta}{2}\right)$$

Control question

[Goldstein, 3.11]

Assume that You scatter a beams of electrons on a thin foil made of heavy ions. The charges, masses and velocities of particles in the beam and foil are given by (q_1, m_1, v_1) and $(q_2, m_2, v_2 = 0)$ respectively. The densities of particle in the beam and the foil is very low, so You may assume that each particle interacts only once with a particle of another species and leaves the beam afterwards. The velocities are not relativistic, so the interaction is dominated by the Coulomb force. Describe how to find the distribution of charged particles (in lab frame) *after* the elastic scattering.

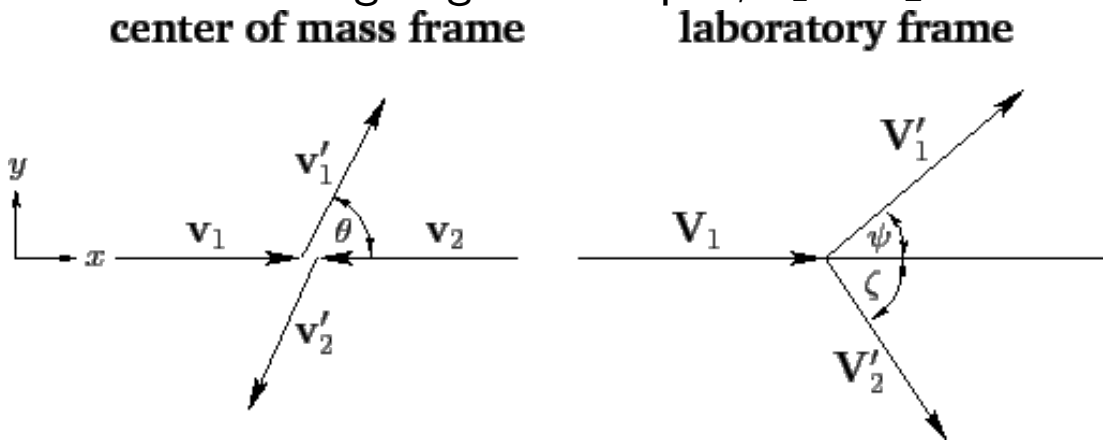
Please suggest how we can solve this problem.

Help:

– What can we say about the center of mass in this problem?

Control question (reply)

- The Coulomb force is central, internal (interaction between particles)
  $\Rightarrow$Center-of-mass will move with constant velocity in lab frame.
- In CM frame:
  – particles move always in opposite directions
  – Relative motion=motion in central field with mass $\mu = m_1 m_2 / (m_1 + m_2)$
  $\Rightarrow$their scattering angles are equal, $\theta_1 = \theta_2 = \theta$



– Scattering angle:

$$\theta_i = \arctan\left(v_y^{(i)} / v_x^{(i)}\right) \qquad (1)$$

– Use Galileo's transform to relate velocities in lab frame and CM frame

$$\boldsymbol{V}_i = \boldsymbol{v}_i + \boldsymbol{V}_{\mathrm{CM}} \qquad (2)$$

$\Rightarrow$Use (1,2) to relate the scattering angles ψ, ζ in CM frame with θ

Control question (reply)

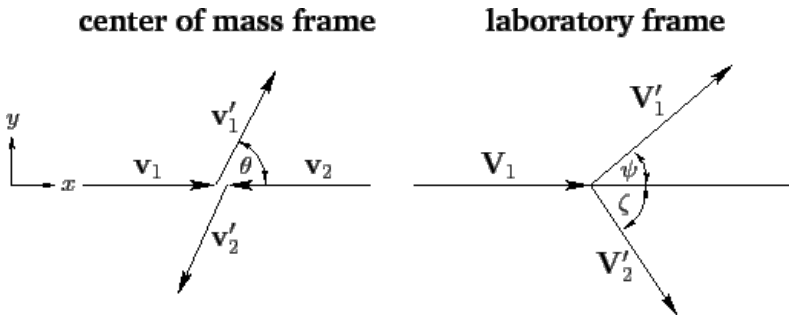
CM velocity:

$$V_{\mathrm{CM}} = \frac{m_1}{m_1 + m_2} v_1 \ \hat{\boldsymbol{x}}$$

Galileo transformation for velocities:

If $\vec{V}_{\mathrm{CM}} \equiv \dot{\vec{R}}_{\mathrm{CM}}$ is the velocity of th center of mass, and $\vec{v}_i$ are the velocities of particles in center-of-mass frame, then in the lab-frame

$$\vec{V}_i = \vec{v}_i + \vec{V}_{\mathrm{CM}}$$



$$\tan\psi = \frac{V_{1y}'}{V_{1x}'} = \frac{v_{1y}'}{v_{1x}' + V_{\mathrm{CM}}} = \frac{v_1 \sin\theta}{v_1 \cos\theta + V_{\mathrm{CM}}}$$

$$\tan\zeta = \frac{V_{2y}'}{V_{2x}'} = \frac{v_{2y}'}{v_{2x}' + V_{\mathrm{CM}}} = \frac{-|v_2| \sin\theta}{-|v_2| \cos\theta + V_{\mathrm{CM}}}$$

$$= \frac{v_1 \sin\theta}{v_1 \cos\theta - (m_2/m_1)\, V_{\mathrm{CM}}}$$

Control question (reply)

Relation between lab-frame angles (ψ, ζ) and the CM frame angle (θ),

$$\tan\psi = \frac{V'_{1y}}{V'_{1x}} = \frac{v'_{1y}}{v'_{1x} + V_{\mathrm{CM}}} = \frac{v_1 \sin\theta}{v_1 \cos\theta + V_{\mathrm{CM}}}$$

$$\tan^2\psi \left(v_1 \cos\theta + V_{\mathrm{CM}}\right)^2 = v_1^2 \left(1 - \cos^2\theta\right)$$

$$\Rightarrow \cos\theta = \cos\psi \sqrt{1 - \tan^2\psi \left(\frac{V_{\mathrm{CM}}^2 + 2 v_1 V_{\mathrm{CM}}}{v_1^2}\right)}$$

–We may relate the cross-sections in the lab-frame and CM frame :

$$d\Omega = 2\pi \sin\theta d\theta, \quad d\Omega_{\mathrm{lab}} = 2\pi \sin\psi \, d\psi$$

$$\frac{d\sigma}{d\Omega_{\mathrm{lab}}} = \frac{d\sigma}{d\Omega} \frac{d\Omega}{d\Omega_{\mathrm{lab}}} = \frac{d\sigma}{d\Omega} \frac{\sin\theta}{\sin\psi} \left|\frac{d\theta}{d\psi}\right| = \frac{d\sigma}{d\Omega} \frac{1}{\sin\psi} \left|\frac{d\cos\theta}{d\psi}\right|$$

Evaluating explicitly the derivative $d(\cos\theta)/d\psi$, we may obtain the cross-section $d\sigma/d\Omega_{\mathrm{lab}}$.

Control question 2

$$E - U\left(r_{\min}\right) - \frac{M^2}{2\,m\,r_{\min}^2} = \frac{m v_\infty^2}{2}\left(1 - \frac{\rho^2}{r_{\min}^2}\right) - U\left(r_{\min}\right) = 0 \qquad (1)$$

– Assume that solving (1), You found more than one root $r_{\min}$. What does this mean physically? Which root should You choose for scattering problem?

Control question 2 (reply)

$$E - U(r_{\min}) - \frac{M^2}{2\,m\,r_{\min}^2} = \frac{mv_\infty^2}{2}\left(1 - \frac{\rho^2}{r_{\min}^2}\right) - U(r_{\min}) = 0 \qquad (1)$$

– Assume that solving (1), You found more than one root $r_{\min}$. What does this mean physically? Which root should You choose for scattering problem?

Reply: This means that potentially we can have bound state orbit with the same energy, which correspond to motion between adjacent roots (r_i, r_{i+1}), assuming energy $E > U_{\text{eff}}$ in that region. In scattering problem particle comes from ∞, so only the largest (outermost) root is relevant.

Control question 3

$$E - U(r_{\min}) - \frac{M^2}{2\,m\,r_{\min}^2} = \frac{m v_\infty^2}{2}\left(1 - \frac{\rho^2}{r_{\min}^2}\right) - U(r_{\min}) = 0 \qquad (1)$$

- We always assumed that the root $r_{\min}$ is simple, so in the vicinity $r \approx r_{\min}$

$$E - U(r) - \frac{M^2}{2\,m\,r^2} \approx \kappa\,(r - r_{\min}) + \mathcal{O}\,(r - r_{\min})^2\,, \quad \kappa = \mathrm{const}$$

In this case we may approximate the integral near $r_{\min}$ as

$$\int_{r_{\min}}^{r} \frac{\frac{M}{m\,r^2}\,dr}{\sqrt{\frac{2}{m}\left(E - U(r) - \frac{M^2}{2\,m\,r^2}\right)}} \approx \frac{M}{r_{\min}^2\sqrt{2\kappa m}} \int_{r_{\min}}^{r} \frac{dr}{\sqrt{r - r_{\min}}}.$$

While the *integrand* diverges, the *integral* converges/is finite.

– Assume that solving (1), for some choice of (E, M) You got double root $r_{\min}$,

$$E - U(r) - \frac{M^2}{2\,m\,r^2} \approx \lambda\,(r - r_{\min})^2 + \mathcal{O}\,(r - r_{\min})^2\,, \quad \lambda = \mathrm{const}$$

so

$$\int_{r_{\min}}^{r} \frac{\frac{M}{m\,r^2}\,dr}{\sqrt{\frac{2}{m}\left(E - U(r) - \frac{M^2}{2\,m\,r^2}\right)}} \approx \frac{M}{r_{\min}^2\sqrt{2\lambda m}} \int_{r_{\min}}^{r} \frac{dr}{|r - r_{\min}|}.$$

has non-integrable singularity, $\phi = \infty$ (?!). What does this mean physically ???

Control question 3 (reply)

–Assume that for some choice of (E, M) You got double root $r_{\min}$,

$$E - U(r) - \frac{M^2}{2\,m\,r^2} \approx \lambda\,(r - r_{\min})^2 + \mathcal{O}\,(r - r_{\min})^2\,, \quad \lambda = \mathrm{const}$$

so

$$\int_{r_{\min}}^{r} \frac{\frac{M}{m\,r^2}\,dr}{\sqrt{\frac{2}{m}\left(E - U(r) - \frac{M^2}{2\,m\,r^2}\right)}} \approx \frac{M}{r_{\min}^2\sqrt{2\lambda m}} \int_{r_{\min}}^{r} \frac{dr}{|r - r_{\min}|}.$$

has non-integrable singularity, $\phi = \infty$ (?!). What does this mean physically ???

Reply: This situation might happen only when energy E coincides with local extrema (maximums) of U_{eff}. In this case $\phi \to \infty$, so particle starts rotating around the center. Note that the same singularity occurs in the expression for $r(t)$,

$$t - t_0 = \int_{r_0}^{r} \frac{dr}{\sqrt{\frac{2}{m}\left(E - U_{\mathrm{eff}}\,(r)\right)}} \sim \int_{r_{\min}}^{r} \frac{dr}{|r - r_{\min}|}$$

so we can see that motion along such trajectory takes infinite time ($t \to \infty$). This special trajectory is called "separatrix".